MILITARY SPECIFICATION

CHARGE, PROPELLING, M218/219/XM220 LOADING, ASSEMBLING AND PACKING

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers requirements and quality assurance provisions for the loading, assembling and packing for the Charge, Propelling, M218/219/XM220 for use on the Cartridge, 81MM; Smoke, RP, M819; Illuminating, M853 and Training, XM879.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1315
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.5)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9300234	- Charge, Propelling, M218
9378134	- Charge, Propelling, M219
9381510	- Charge, Propelling, XM220
9313721	- Packing and Marking for Box, Fiberboard, Outer for Charge, Propelling, M205 or Container Assembly
9313756	- Packing for Pallet for Box, Fiberboard, Outer for Charge Propelling M205 or Container Assembly

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Assembly. The propelling charge assembly shall comply with all requirements specified on drawing 9300234/9378134/9381510 and associated drawings and with all requirements specified in applicable specifications and standards.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

3.3 Gross weight of propelling charge The weight of the loaded propelling charge shall not vary more than plus or minus eleven grains from the assessed propellant weight plus the median drawing weight of the container assembly when tested as specified in 4.5.2.

3.4 Velocity. The propelling charges shall impart a corrected muzzle velocity (See 6.9) between the listed limits as follows:

Cartridge	Prop Chg.	Velocity (fps) Minimum	Maximum
M819	M218	910	940
M853	M219	1005	1035
XM879	XM220	990	1020

The propelling charges shall be tested as specified in 4.5.3. The standard deviation of the lot shall not be greater than 6.5 fps.

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship.

3.6.1 Propelling charge. The propelling charge shall be free of dirt, grease or other foreign material.

3.6.2 Packaging. The packaging shall be free from foreign material. The boxes shall be assembled without undue binding and no flap shall project beyond the edge of the box when closed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items (as applicable) in sample quantities as indicated.

Part Description	Drawing	Quantity
Charge, Propelling, M218	9300234	210
Charge, Propelling, M219	9378134	210
Charge, Propelling, XM220	9381510	210

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078(AR).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70545(AR)

PARAGRAPH	TITLE: Charge, Propelling, M218/219/XM220		SHEET 1 OF 1		DRAWING NUMBER: 9300234/9378134/9381510
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for defects	110		3.2	4.4.2.1
	Weight of propellant	10		3.2	4.5.2
	Gross weight of propelling charge	110		3.3	4.5.3
	Velocity (Proving Ground)	*200		3.4	4.4.3.1/4.5.4

NOTES:

*200 is equivalent to the max. charge of 50 cartridges (4 charges per cartridge).

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of propelling charges shall contain:

a. Container parts of one lot interfix from one manufacturer.

b. Propellant from not more than one lot, from one manufacturer.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70545(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Charge, Propelling, M218/219/XM220	1 OF 1	9300234/9378134 9381510
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Weight of propellant in excess of 10 grains below min.	1/	100%	3.2	4.5.1
2	Gross weight of propelling charge, min.	2/	100%	3.3	4.5.3
3	Container or container seal damaged to the extent that propellant shall escape		100%	3.2	Visual (4.5.5)
Major					
101	Weight of propellant from min. to and including 10 grains below min.	1/	100%	3.2	4.5.1
102	Weight of propellant, max.	1/	100%	3.2	4.5.1
103	Gross weight of propelling charge, max.	2/	100%	3.3	4.5.3
104	Velocity (Proving Ground)	200/100 3/	1.0%	3.4	4.4.3.1/4.5.4
105	Container or container seal damaged but not to the extent that propellant shall escape		100%	3.2	Visual (4.5.5)
Minor					
201	Evidence of poor workmanship		0.65%	3.6.1	Visual

NOTES: 1/ This inspection may be performed prior to loading. See 4.5.1 for selection of test method.

2/ In the automated mode, this weight is the total loaded container weight minus the empty container weight for each loaded increment. 3/ 200/100 is equivalent to the max. charge of 50/25 cartridges (4 charges per cartridge).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

MIL-C-70545(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Packing and Marking for Box, Fiberboard, Outer for Charge, Propelling, M205 or Container Assy.	1 OF 1	9313721
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Charges incorrectly packed		0.40%	5.1	Visual
102	Barrier bag damaged (punctured)		0.40%	5.1	Visual
103	Marking improper (barrier bag, outer box)		0.40%	5.3	Visual
104	Improper seal (barrier bag, inner and outer boxes)		0.40%	5.1	Visual
105	Orientation stripe on corner of outside box improperly located (with respect to loading holes of charges)		0.40%	5.1	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6.2	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70545(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Packing for Pallet				9313756
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Orientation of boxes improper		0.40%	5.2	Visual
102	Strapping loose		0.40%	5.2	Visual
103	Boxes damaged		0.40%	5.2	Visual
Minor	None defined				

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

4.4.3 Testing.

4.4.3.1 Velocity.

4.4.3.1.1 First article. The sampling plan for cartridges required and procedure/acceptance criteria for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter M and an AQL of 1.0 percent. The test shall be performed as specified in 4.5.4.

4.4.3.1.2 Production.

4.4.3.1.2.1 Initial production. Beginning with the first lot produced and continuing until three consecutive lots of propelling charges have complied with the applicable requirements, the sampling plan and required procedures for cartridges for this test shall be in accordance with MIL-STD-414 Section B, Table B-3, Code Letter M and an AQL of 1.0 percent. The test shall be performed as specified in 4.5.4.

4.4.3.1.2.2 Subsequent production. After three consecutive lots have met the criteria of 4.4.3.1.2.1 the sampling plan for cartridges for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter I, and an AQL of 1.0 percent. The test shall be performed as specified in 4.5.4. In the event of a lot rejection the sampling plan of 4.4.3.1.2.1 shall be required beginning with the next lot and continuing until three consecutive lots of propelling charges have complied with the applicable requirements.

4.5 Methods of inspection.

4.5.1 Weight of propellant.

4.5.1.1 Manual method.

4.5.1.1.1 Method 1. The propellant shall be weighed. Weight limits for weighing should be established within the applicable weight requirements, depending upon the accuracy and precision of the weighing system to assure compliance with the charge weight requirements. Any charge of bulk propellant failing to meet the applicable requirement shall be classified defective and removed from the lot.

4.5.1.1.2 Method 2. Samples for this test shall be selected from propelling charges that have met the requirements of 3.3. A sample of 200 propelling charges shall be randomly selected during production for each shift. The propellant shall be removed from the charge, and weighed. If one or more critical defects or two or more major defects are found, the production represented by the sample shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

4.5.1.2 Automated Method.

4.5.1.2.1 Method 1. The contractor may elect to perform this operation with automated equipment and mandatory process controls. The automated equipment shall weigh the container and place this weight in memory and then weigh/checkweigh the propellant before loading. After loading, the total weight of the container/ propellant combination shall be weighed to verify complete loading by subtracting the increment weight (from memory) from the total weight to obtain the loaded propellant weight. In such cases, process control plans in accordance with Data Item Description DI-P-1604 must be submitted to the Technical Agency (see 6.6) for review of the automated operation.

4.5.1.2.2 Method 2. The empty increments shall be sorted into discreet weight groups with a minimum group range of 0.6 grains to minimize container weight variance. After the empty increments are sorted, they will be loaded by groups. The automated equipment shall weigh and check weigh the propellant before loading. After loading, the total weight of the container and propellant combination shall be weighed to verify complete loading by subtracting the increment group weight from the total weight to obtain the loaded propellant weight.

4.5.2 Weight of propellant in charge, propelling (first article).

4.5.2.1 Destructive test. The propellant shall be removed from the charge, propelling and weighed. If any sample fails to meet the requirement it shall be classified defective and the first article sample shall be rejected.

4.5.2.2 Non-destructive (alternate procedure). The charge, propelling shall be weighed and the net weight of propellant determined by subtracting parts tare weight from the gross weight. If any sample fails to meet the requirement it shall be classified defective and the first article sample shall be rejected.

4.5.3 Gross weight of propelling charge. The loaded propelling charge shall be weighed individually. If any assembly fails to meet the requirement, it shall be classified defective and removed from the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

4.5.4 Velocity test. This test shall be conducted at a Government proving ground. The rounds shall be inert loaded to an as fired weight of as follows:

Cartridge	Prop Chg.	Rd Weight
M819	M218	10.35 ± .05 pounds
M853	M219	8.90 ± .05 pounds
XM879	XM220	9.00 ± .05 pounds

The weight shall include the dummy fuze. The rounds shall be assembled at maximum charge (i.e. with four charges). The rounds shall be conditioned to 70° ± 2.5° Fahrenheit and fired from a vehicle mount at an elevation of 800 mils, from an 81MM M253 mortar tube. The individual velocities and pressures shall be obtained and recorded. Velocity standard deviation and mean velocity shall be calculated to determine compliance with applicable requirements. A formal firing report shall be submitted (see 6.7).

4.5.5 Damaged propelling charge. To ascertain that a damaged propelling charge is to be classed as critical or major, the following procedures shall be used:

a. Place the damaged charge on an M24/28/29 fin, oriented such that the propellant would be resting against the damaged surface.

b. The fin-charge assembly shall be hand-shaken vigorously three times.

c. Examination shall be made for propellant spillage - if any propellant is observed, the damaged propelling charge shall be classed as a critical defect; if no propellant is observed, the damaged propelling charge shall be classed as a major defect.

5. PACKAGING

5.1 Unit packing. Packing shall be as specified on dwg. 9313721.

5.2 Exterior packing. Packing shall be as specified on dwg. 9313756.

5.3 Marking. Marking shall be as specified on dwg. 9313721 and dwg. 9313756.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the 81MM Mortar for use on the Cartridge, 81MM, Smoke RP, M819; Illuminating, M853; and Training, XM879.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army Armament, Research, Development and Engineering Center, ATTN: AMSMC-QAT-I(D), Dover, NJ 07801-5001.

6.4 Data requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data Requirement	Applicable DID
4.5.1.2	Manufacturing Methods Reports	DI-P-1604

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as ARRADCOM, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of Armament Research and Development Center (ARDC).

6.6 Submission of process control plans for approval. Submit plans as required to Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-B(D), Dover, NJ 07801-5001.

6.7 Submission of test data. Formal firing reports containing velocity test data and lot acceptability shall be sent to Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-B(D), Dover, NJ 07801-5001.

6.8 Proving ground test summary.

Test	Sample Size	Requirements
First Article/ Velocity	200 Propelling Charges (M218/219/XM220) (50) 81MM M819/853/XM879 Inert Cartridge and Fuzes w/Acceptable Ignition Cartridge and (17) Calibration Cartridges	See Paragraph 3.4

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

First Three Production Lots/ Velocity	200 Propelling Charges (M218/219/XM220) (50) 81MM M819/853/XM879 Inert Cartridge and Fuzes w/Acceptable Ignition Cartridge and (17) Calibration Cartridges	See Paragraph 3.4
Subsequent Production/ Velocity	100 Propelling Charges (M218/219/XM220) (25) 81MM M819/853/XM879 Inert Cartridge and Fuzes w/Acceptable Ignition Cartridge and (9) Calibration Cartridges	See Paragraph 3.4

6.9 Velocity requirement. The velocity requirement in this specification is established for testing mortar cartridges fired from a hard mount (armor plate). Since all velocity assessments will be conducted from hard mounts, a correction factor applied to the assessed velocities will reflect the appropriate firing table velocities when a standard mortar base plate is used.

6.10 Subject term (key word) listing.

Ammunition, mortar
Charge, propelling
Velocity, muzzle
Velocity, standard deviation
Weight, gross
Weight, propellant

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-A677)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-70545	CHARGE, PROPELLING, M218/219/XM@@) LOADING, ASSEMBLING AND PACKING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that the content is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)